Check for updates

Received: 7 October 2021 | Revised: 26 December 2021 | Accepted: 24 January 2022

DOI: 10.1111/1440-1681.13634

CEPP Clinical and Experimental Pharmacology and Physiology WILEY

ORIGINAL ARTICLE

Dual-target synergistic antithrombotic mechanism of a dabigatran etexilate analogue (HY023016)

Sheng Wan[1] | Shuang Cui[1] | Minrui Jiang[1] | Qian Wu[1] | Yingying Ji[1] | Yungen Xu[2] | Guoqing Gong[1]

[1]Department of Pharmacology, China Pharmaceutical University, Nanjing, China

[2]Department of Medicinal Chemistry, China Pharmaceutical University, Nanjing, China

Correspondence
Guoqing Gong and Yungen Xu, China Pharmaceutical University, 639 Longmian Avenue, Jiangning District, Nanjing 211198, China.
Emails: gonggq@hotmail.com; xu_yungen@hotmail.com

Present address
Shuang Cui, State Key Laboratory of Natural Medicines, Key Lab of Drug Metabolism and Pharmacokinetics, China Pharmaceutical University, Nanjing, China

Funding information
National Natural Science Foundation of China, Grant/Award Number: 81373415 and 21272277

Abstract

Thrombin has long been considered a desirable antithrombotic target, but anti-thrombin therapy without anti-platelet therapy has never achieved the ideal effect. HY023016, derived from dabigatran etexilate, exhibited a potent antithrombotic efficacy. In the present study, mechanisms underlying this effect were explored. HY023016 strongly decreased the binding of thrombin to recombinant GPIbα N-terminal sequence, which was confirmed by surface plasmon resonance. Flow cytometry revealed that HY023016 selectively decreased the binding of antibody to GPIbα and inhibited the washed human platelet aggregation induced by thrombin. Fluorescence experiment showed that HY023016 remarkably inhibited exosite II by a loss of affinity for the γ'-peptide of fibrinogen. Using intravital microscopy, we observed and recorded the dynamic process of thrombus formation and found that HY023016 effectively prevented thrombus formation in rat arteriovenous shunt thrombosis model. On the basis of these findings, we propose that HY023016 provides a novel insight into the antithrombotic mechanism, which exerts synergistic anticoagulant and antiplatelet effects through thrombin and GPIbα.

KEYWORDS
dual-target, GPIbα, platelet membrane glycoproteins, thrombin, thrombin exosite II

1 | INTRODUCTION

Currently, the single target inhibitors—direct thrombin inhibitors (dabigatran etexilate and AZD0837), factor Xa inhibitor (apixaban and rivaroxaban), platelet P2Y12 receptors antagonist (clopidogrel and ticagrelor), platelet integrin αIIbβ3 inhibitor (abciximab and tirofiban), etc.—are used in the clinical antithrombotic therapies.[1,2] Traditional Chinese medicines have also been widely applied in cardiovascular diseases, such as tetramethylpyrazine.[3,4] Wu et al[5] reported that tetramethylpyrazine inhibited GPIbα to attenuate platelet aggregation and prevent thrombus formation. However, because of the limitations of current treatments, their application is restricted to one type of venous or arterial thrombotic. These drugs are also accompanied by adverse reactions: single anti-thrombin therapy without any complementary antiplatelet therapies could increase the level of platelet activation and elevate the risk of myocardial infarction,[6] whereas excessive bleeding caused by thrombocytopenia and unexpected platelet activation occurs in individual antiplatelet therapy.[2,7] For this reason, there is an urgent need to design novel anti-thrombotic agents to improve anti-total thrombotic risk efficacy.

We designed and synthesized a series of new compounds from the core of dabigatran etexilate (direct thrombin inhibitor) and the active groups of tetramethylpyrazine (with antiplatelet effect) by combinatorial chemistry. We conducted pharmacodynamic screening of the drugs, and HY023016 with good antithrombotic activity was discovered and selected for further study (Figure 1).

Thrombin is vital for the coagulation cascade,[8] and for highly specific interactions and regulatory functions of thrombin, allosterism is indispensable.[9] In the previous studies, we have demonstrated

 © 2022 John Wiley & Sons Australia, Ltd

that HY023016 could inhibit thrombin active site and the antagonistic activity was slightly stronger than dabigatran etexilate at the same molarity.[10] Before the present study, we have discovered that HY023016 resulted in a significant loss of affinity for exosite II ligand by heparin-sepharose chromatography in preliminary experiments. The position of exosite II subdomain may disorder the active site, so we expected that the active sites of thrombin and exosites were compromised when they were inhibited by HY023016 to present an energetically anticoagulant effect. Remarkably, we further found that HY023016 could attenuate the interaction between GPIbα and von Willebrand factor (vWF), which mediates platelet adhesion, activation and aggregation under high flow conditions.[11] Interestingly, GPIbα amino terminal fragments contain the binding sites for thrombin and vWF.[12,13] The binding site for vWF-A1 is three sulphated Tyr residues (Tys276, Tys278 and Tys279) position the GPIbα sequence, which is corresponding to thrombin.[14] Therefore, we hypothesized

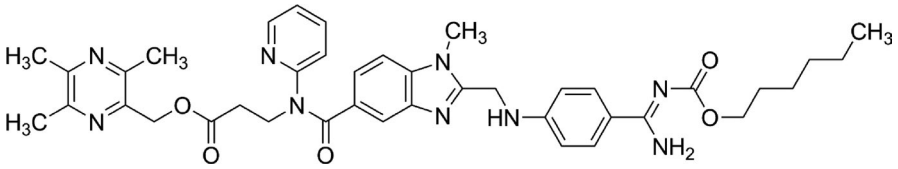


FIGURE 1 Chemical structure of HY023016

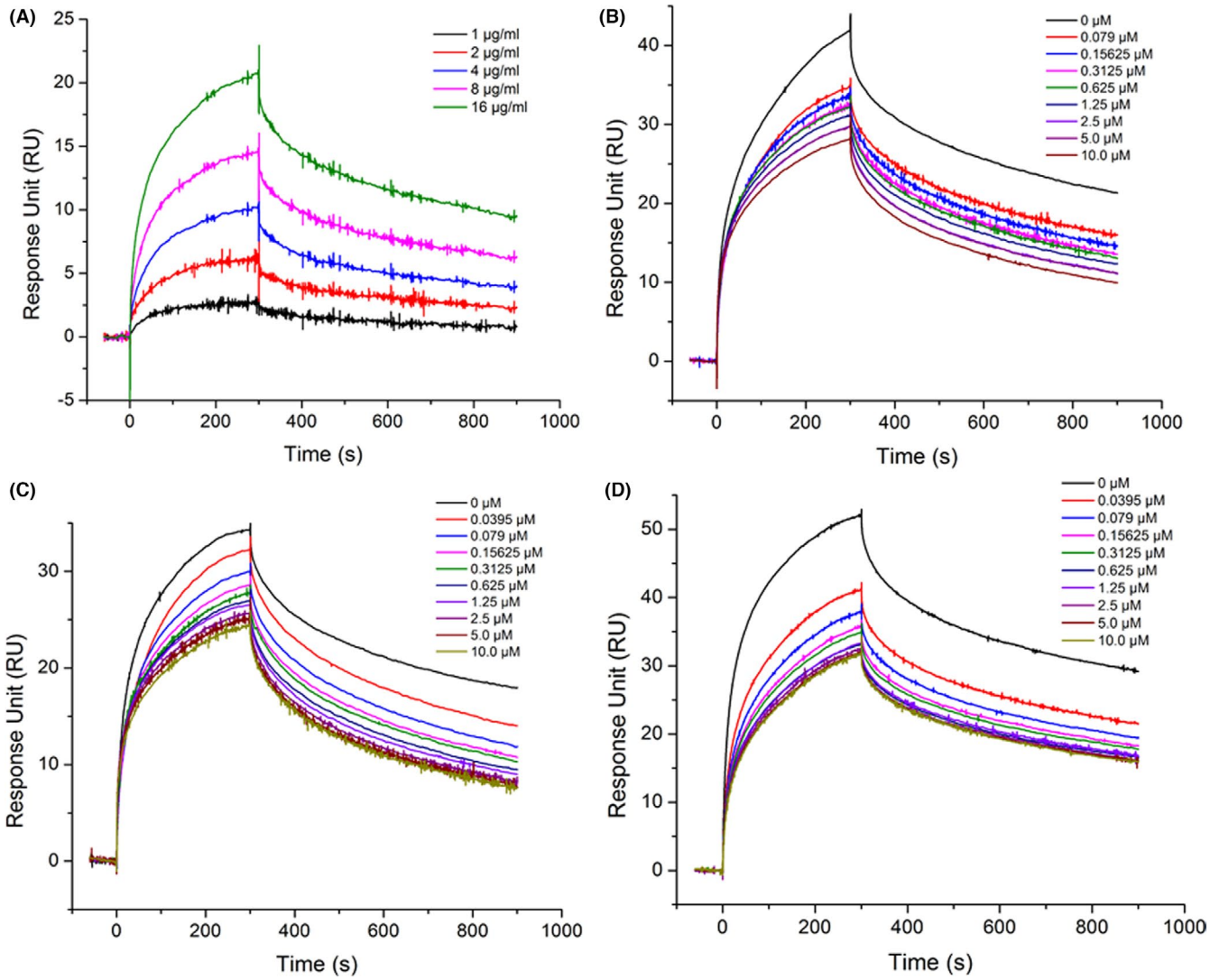


FIGURE 2 Binding of GPIbα 1-300 aa to immobilized thrombin. (A) Increasing concentrations of GPIbα 1-300 aa up 16 μg/ml were injected at a rate of 30 μl/min into flow cells of a CM5 biosensor chip containing immobilized thrombin. The data were analyzed by nonlinear regression to a rectangular hyperbola to determine Kd. GPIbα (50 μg/ml) was injected into flow cells containing immobilized thrombin in the presence of 0–10 μM HY023016 (B), dabigatran etexilate (C), or tetramethylpyrazine (D). Response units (RU) are plotted versus time for the indicated different test compound concentrations

14401681, 2022, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/1440-1681.13634 by Harvard University, Wiley Online Library on [09/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

that HY023016 may exert anticoagulation role via thrombin exosite II in addition to active site and may also exert antiplatelet effect because of the structural modification group.

In this article, the additive anticoagulant and antiplatelet action of HY023016 was evaluated. As a dual-target drug, HY023016 is fundamentally different from the current clinically used agent. We attempted to highlight the importance of multiple-component, multiple-target drugs in a new conceptual framework for antithrombosis therapy to reduce the 'total thrombotic risk'.[15,16] Taken together, we anticipate that HY023016 could inspire new ideas and research to further bring antithrombotic agents with potential dual anticoagulant and antiplatelet properties into the clinic.

2 | RESULTS

2.1 | HY023016 inhibited GPIb-VWF interaction in vitro

Previously, we expressed human GPIbα N-terminal 1-300 amino acids containing the GPIbα-thrombin binding site in mammalian HEK293 cells by establishing the GPIbα-His overexpression plasmid and analysed its purity over 90% by SDS-Page analysis (Figure S1). Thrombin was allowed avid binding to GPIbα as demonstrated by surface plasmon resonance. Increasing concentrations of purified GPIbα 1-300 aa were injected over a CM5 sensor chip loaded with α-thrombin (1500 RU). GPIbα peptide bound to the immobilized thrombin with a K_d of 25.73 nM (Figure 2A). Subsequently, thrombin binding to the GPIbα was measured in the presence of increasing concentrations of HY023016, dabigatran etexilate and tetramethylpyrazine (Figure 2B-D). All of them reduced thrombin binding to GPIbα in a concentration-dependent manner. HY023016 showed a comparable inhibitory activity compared with tetramethylpyrazine and the IC_{50} were 26.33 ± 8.19 and 36.38 ± 9.02 μM, respectively. Both exerted much stronger inhibitory effect than dabigatran etexilate with IC_{50} values of 129.2 ± 46.66 nM ($P < .01$).

2.2 | HY023016 competitive binding to GPIbα receptors

To evaluate the role of HY023016 on platelet surface protein, using flow cytometry we analysed the antibody binding rate of three proteins: GPIbα (CD42b), GPIIIα (CD61) and P-selectin (CD62P), which are involved in platelet activation. We drew the positive fluorescence intensity range gate in histogram and count the percentage. As shown in Figure 3A, in the presence of 20 μM HY023016, the anti-human GPIbα fluorescence intensity was significantly lower than that in the absence of HY023016 (61.40% ± 2.09% vs 77.30% ± 2.55%). HY023016 concentration dependently weakened the binding of the antibody to GPIba by competitive combination with GPIba with a calculated IC_{50} of 129.09 ± 48.50 μM. In contrast, HY023016 at the same concentrations did not induce binding rate change of GPIIIα and P-selectin (Figure 3B,C).

2.3 | HY023016 disrupted thrombin binding to fibrinogen γ'-peptide

To test the effects of HY023016 on exosite II of thrombin, we conducted a fluorescence experiment. Before the competitive tests, thrombin active site was blocked with saturating PPACK at the stoichiometric ratio of 100: 1 (α-thrombin: PPACK). As shown in Figure 3D, a large fluorescence quench was elicited at the emission peak of 528 nm (from 574 to 409) after addition of PPACK-thrombin. As expected, when it was added to the cuvette, HY023016 caused a reduction in quench, suggesting that the interaction between the exosite II and γ' -peptide had been significantly diminished. HY023016 decreased affinity for γ'-peptide and exosite II in a concentration-dependent manner (Figure 3E), with an IC_{50} value of 0.77 ± 1.30 μM.

To further evaluate the inhibitory effect of HY023016 on exosite II, we assessed K_d values of α-thrombin for γ'-peptide by fluorescence titration in the presence and the absence of HY023016. As shown in Figure 3F, α-thrombin bound γ'-peptide with K_d values of 0.18 ± 0.04 μM. In the presence of 400 nM of HY023016, the K_d value was quantified. The affinity of thrombin for γ'-peptide significantly decreased when HY023016 was added (K_d = 1.93 ± 0.23 μM, $P < .01$). Titrations using recombinant thrombin, R93E, gave a K_d value of 2.28 ± 0.28 μM, which resulted in an ~10-fold loss compared with wild type thrombin.

2.4 | HY023016 inhibited thrombin-induced platelet activation in vitro

To analyse the effect of HY023016 on the platelet activation, thrombin-induced platelets were assessed for PAC-1 and P-selectin surface content using flow cytometry. Figure 4 shows that the estimated expression of PAC-1 and P-selectin significantly increased by ~5-7 fold ($P < .01$) and ~4-5 fold ($P < .01$), respectively, in the presence of thrombin compared with that in the absence of thrombin. Clearly, HY023016, dabigatran etexilate and tetramethylpyrazine concentration dependently downregulated the surface content of PAC-1 and P-selectin in human gel-filtered platelets (GFP) with calculated IC50 values for HY023016: 6.83 ± 1.20 and 2.80 ± 1.34 μM (Figure 4A), respectively, and for dabigatran etexilate: 7.85 ± 1.44 and 1.91 ± 1.26 μM (Figure 4B), respectively, and for tetramethylpyrazine: 8.13 ± 1.23 and 2.73 ± 1.39 μM (Figure 4C), respectively.

2.5 | HY023016 inhibited rat arteriovenous shunt thrombus formation in vivo

To investigate the effect of HY023016 on thrombus formation in vivo, we used rat arteriovenous shunt thrombosis model in real-time 15 minutes after treatment by an intravital microscopy. As shown in Figure 5A, several minutes after shunt was opened, platelets started to aggregate, formed and grew into thrombi on the silk thread. In

14401681, 2022, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/1440-1681.13634 by Harvard University, Wiley Online Library on [09/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

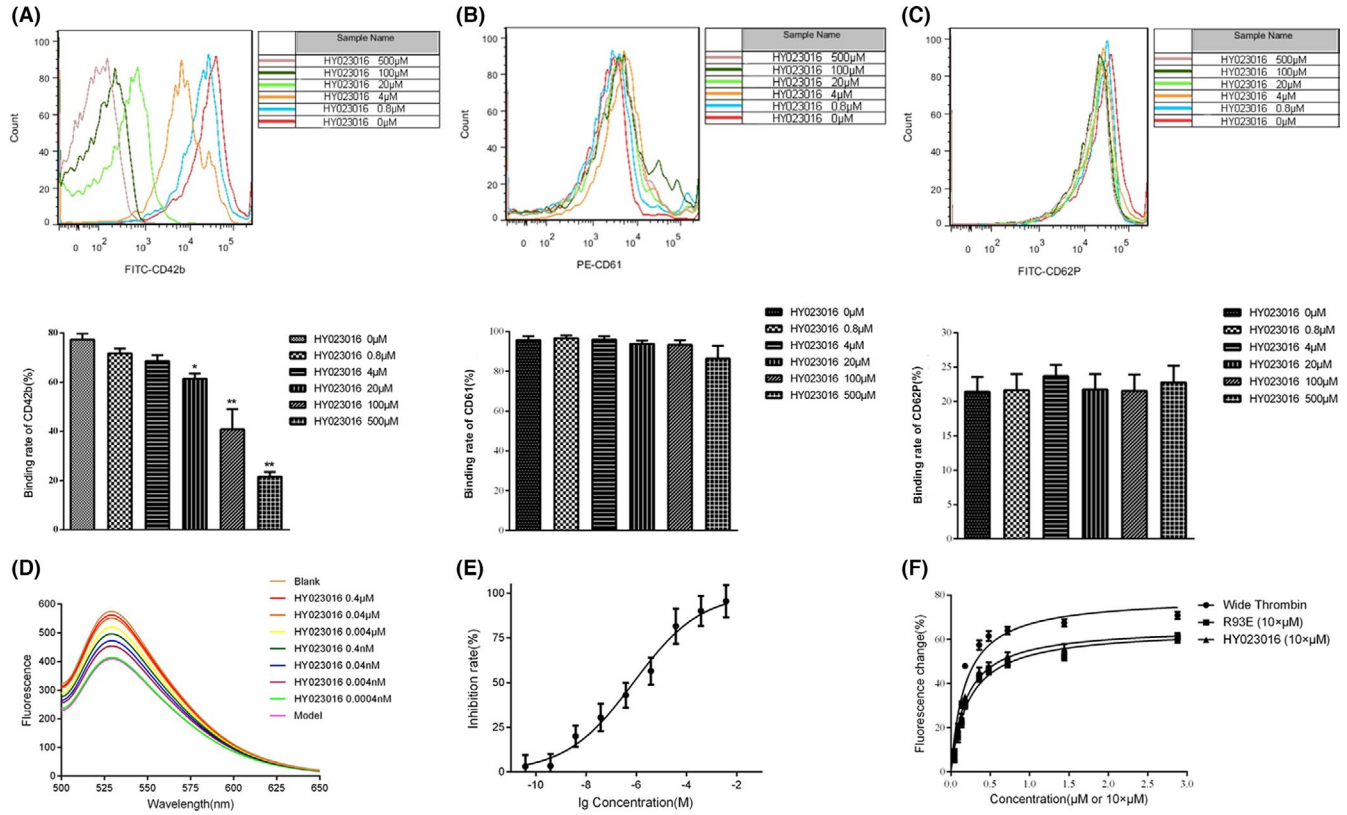


FIGURE 3 The interaction sites of HY023016 on platelet and thrombin. Washed platelet (100 μL) with vehicle or HY023016 was incubated at 37°C for 30 min. Then, FITC-CD42b (A), PE-CD61 (B) or FITC-CD62P (C) was added in the mixture. After 15 min incubation at room temperature, the reaction was terminated by 1% paraformaldehyde (300 mL) and the samples analysed by flow cytometry (n = 3). D. The fluorescence spectra of fluorescein-labeled fibrinogen γ'-peptide with PPACK-thrombin inhibited by HY023016 (0-0.4 μM). E. Inhibitory effect of HY023016 on thrombin exosite II (n = 5). F. Titration of wide PPACK-thrombin (circles), PPACK-thrombin inhibited by HY023016 (squares), and R93E (triangle) into solutions containing the fluorescein-γ'-peptide, with non-linear fits as lines. The concentration of PPACK-thrombin is μM units, whereas the concentration of R93E and PPACK-thrombin inhibited by HY023016 are 10 times higher than indicated (n = 3)

vehicle rats, visible platelets formed multiple emboli that were more than 20 μm in diameter at the fifth minute. At 15 minutes the shunt was partially blocked by stable thrombus. In contrast, treatment with HY023016 significantly inhibited the process regardless of the initial platelet interaction or the final stabilizing thrombi formation. Further statistical analysis revealed that the time to form the first thrombus (>20 μm) was significantly prolonged in the treatment with 5 mg/kg of HY023016 (593.8 ± 207.7 seconds) as compared to vehicle rats (135.5 ± 37.55 seconds) (Figure 5B). The number of emboli (>20 μm) in 15 minutes was markedly decreased by 2.5 mg/kg of HY023016 (33.33 ± 8.16), whereas the control was 59.17 ± 9.85, $P < .01$) (Figure 5C). Moreover, HY023016 inhibited the dynamic formation of thrombus in a dose-dependent manner. We also compared the anti-thrombotic effect of HY023016, dabigatran etexilate and tetramethylpyrazine. At the same concentration, all of them similarly prevented rat arteriovenous shunt thrombosis over 15 minutes, including significant prolongation of thrombus formation time and reduction of the number of emboli.

Moreover, we measured the wet thrombus weight after extracorporeal circulation was maintained for 15 minutes. Compared with the vehicle control, HY023016 significantly decreased the wet thrombus weight from 9.01 ± 2.43 to 5.750 ± 1.79 mg, 4.62 ± 1.27 and 3.80 ± 1.67 mg at doses of 2.5, 5, and 10 mg/kg, respectively (Figure 5D). Thrombus weight was substantially reduced by dabigatran etexilate (4.67 ± 1.71 mg) and tetramethylpyrazine (5.52 ± 1.86 mg). We also found HY023016 robustly prolonged prothrombin time (PT) and activated partial thromboplastin time (aPTT) in a dose-dependent manner (Figure 5E,F).

3 | DISCUSSION

Here, we assessed the anti-thrombotic mechanisms of HY023016 both in vivo and in vitro. HY023016 provided a new pattern with regard to anti-thrombotic mechanism.

Crystal structure of GPIbα amino terminal fragments with thrombin has certified that the sulphate groups on Tyr^{276}, Tyr^{278} and Tyr^{279} of GPIbα are essential positions for binding.[17,18] We constructed GPIbα extracellular 1-300 amino acid sequence to estimate the inhibitory effect of HY023016 on GPIbα binding to thrombin. Considering that GPIbα is a heavily glycosylated secretory protein, a carbohydrate ratio ~60% of total molecular weight, we applied a mammalian expression system capable of posttranslational modification.

14401681, 2022, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/1440-1681.13634 by Harvard University, Wiley Online Library on [09/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

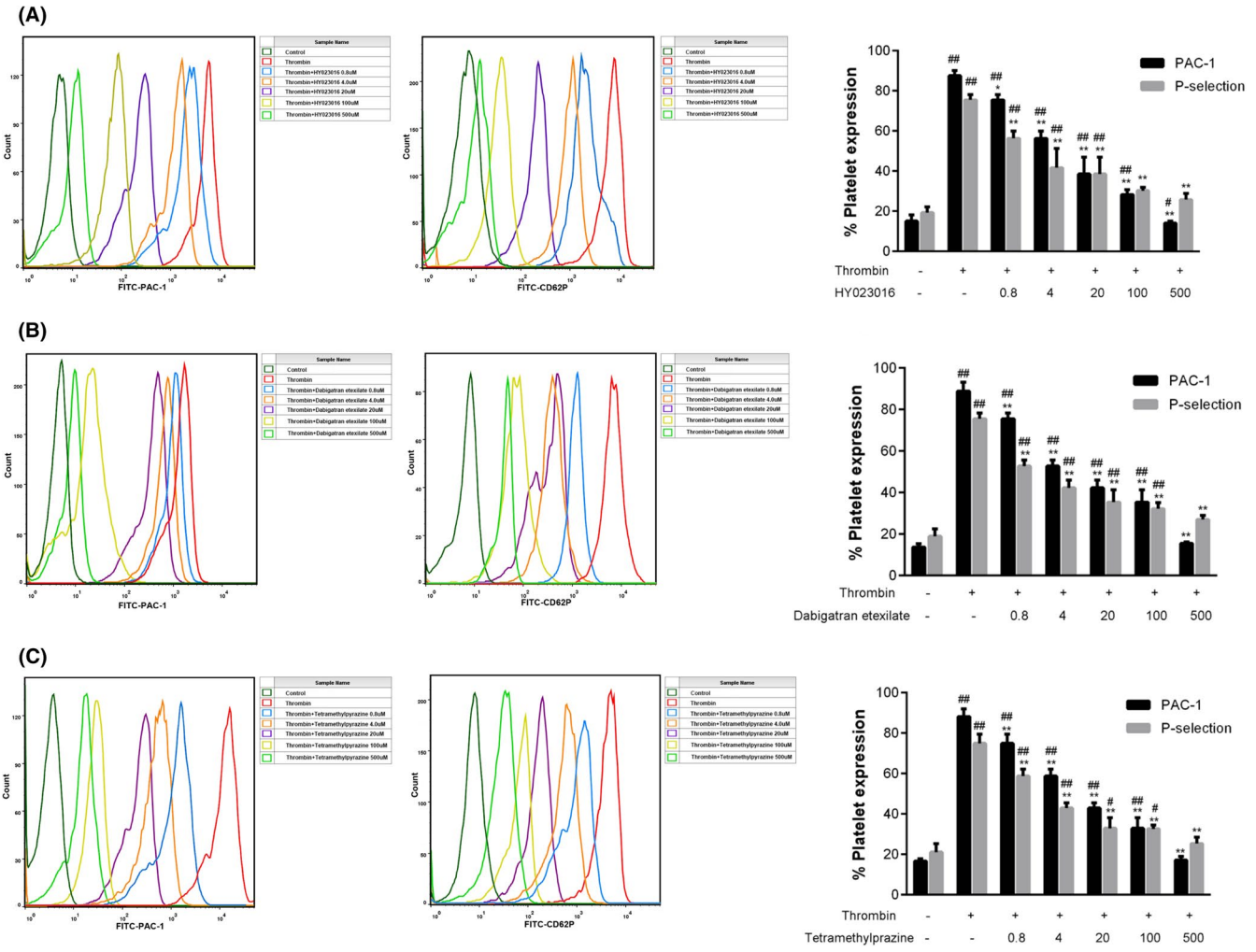


FIGURE 4 Effect of HY023016 on thrombin induced platelet aggregation. In each case, the black column is PAC-1 expression and the gray column is P-selectin. washed platelet was incubated with 0-500μM HY023016 (A), dabigatran etexilate (B) or tetramethylpyrazine (C) at 37°C for 30 min, and then FITC- PAC-1 or anti-P-selectin was added, respectively. The samples analysed by flow cytometry (n = 3). Data are expressed as mean ± SD. *P<0.05 vs control, **P<0.01 vs control; #P<0.05 vs blank, ##P<0.01 vs blank

Surface plasmon resonance (SPR) data manifest that the interaction between GPIbα and thrombin is a dynamic process of rapid binding and slow dissociation. Consistent with our assumptions, HY023016 showed a slightly better inhibitory activity than tetramethylpyrazine despite no significant difference. This may be related to the synergistic effect of HY023016 on double targets. Interestingly, GPIbα binding to thrombin was also suppressed in the presence of dabigatran etexilate. This phenomenon may be attributed to a complex allosteric network within thrombin. The thrombin substrate specificity depends on the positively charged domains flanking the active site, namely thrombin external sites I and II.[19,20] These two external sites modulate thrombin activity through steric hindrance by binding to inhibitors or cofactors, while also altering thrombin activity by allosteric action. Concretely, active site combined with dabigatran etexilate perturbed the binding of exosite II to GPIbα. The inhibitory effect of dabigatran etexilate, however, was far less than that of HY023016 and tetramethylpyrazine.

In terms of the platelet site of HY023016, we analysed several major protein receptor subunits of platelet membrane surface, including GPIbα, GPIIIa and P-selectin. As shown by flow cytometry, HY023016 was unable to induce a decrease in GPIIIa and P-selectin levels, yet it specifically reduced GPIbα expression on platelet surface, suggesting that HY023016 selectively binds to GPIbα. GPIbα, an important functional site on GPIb-IX-V complex, and involves in the binding of platelets to vWF[21] and thrombin. It plays a pivotal role in both haemostasis and thrombosis. This may be another reason that HY023016 is superior to dabigatran etexilate for inhibition of the binding of thrombin to GPIbα.

There are several ligands that bind to thrombin exoprotein II, including the γ'-peptide of fibrinogen, heparin and HD22. Therefore, fluorescein-labelled γ'-peptide of fibrinogen was created to identify the effect of HY023016 on exosite II of thrombin. Many studies have shown that ligands binding to the active site of thrombin leads to allosteric changes at the exosites.[22,23] To exclude the false positive

14401681, 2022, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/1440-1681.13634 by Harvard University, Wiley Online Library on [09/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

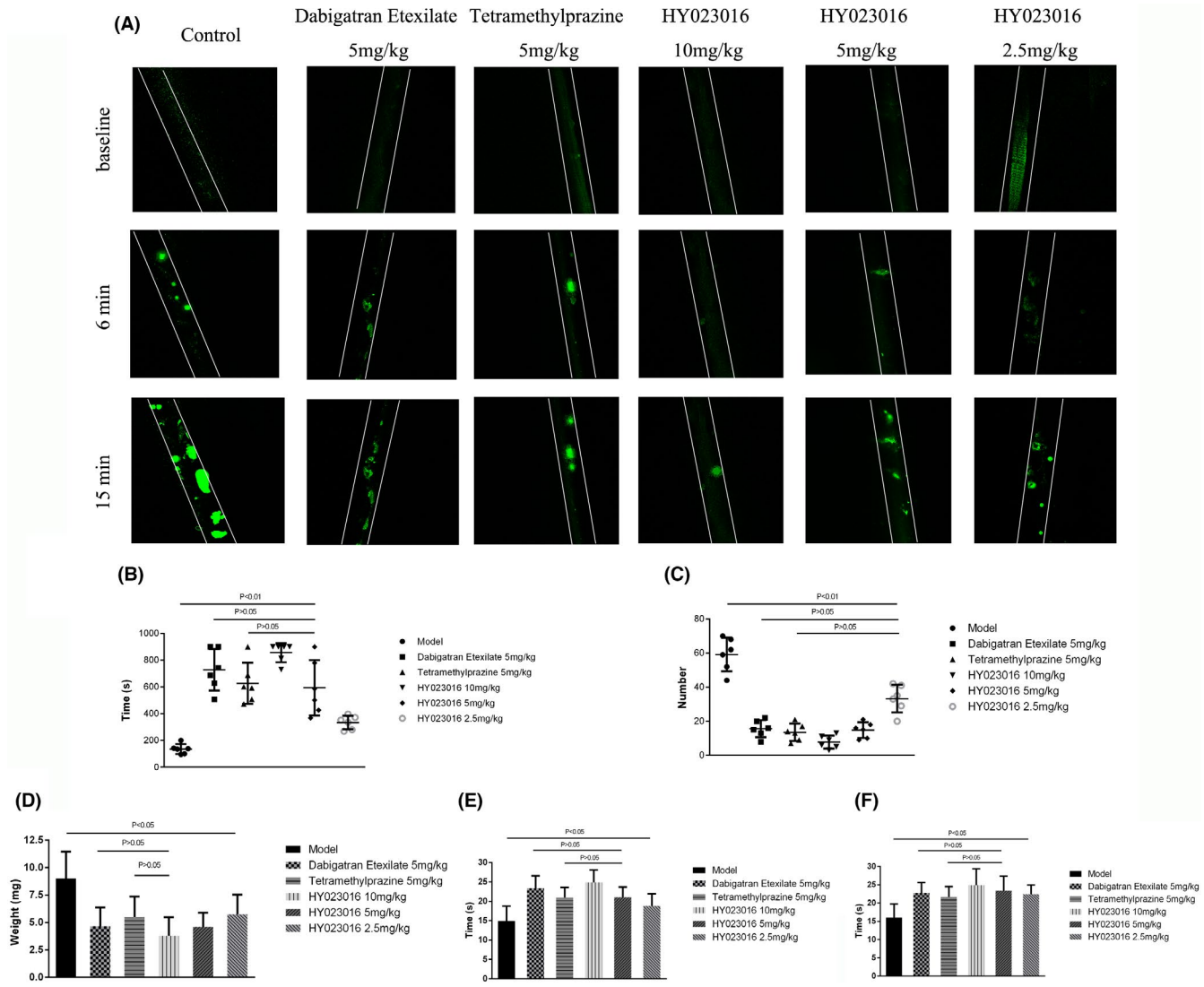


FIGURE 5 HY023016 inhibits rat arteriovenous shunt thrombosis. (A) Representative images of thrombus formation at baseline, 6 and 15 min after shunt was open. Blood flow is from top to bottom. The time to first thrombus (>20 μm;) (B) number of emboli (>20 μm) in 15 min, (C) in the pipeline of untreated and treated mice were determined. Thrombosis weight(D), PT(E) and (aPTT) (F) were measured after 15 min observation. Data are expressed as mean ± SD (n = 6)

results caused by the binding of HY023016 to thrombin active site, we used PPACK to block thrombin active site that does not induce functional changes of exosites. IC_{50} values and dissociation constants were determined. In the presence of HY023016, thrombin exhibited a significant loss of affinity for γ'-peptide. According to published crystal structures, Arg93 is the principal site for the exosite II-mediated the interactions of thrombin with ligand, and others include Arg101, Arg126, Arg233, Lys235, Lys236 and Lys240. Likewise, R93E, charge reversal mutation within exosites II, showed a slight affinity for γ'-peptide, comparable to the presence of HY023016.

Platelet activation and aggregation are essential for primary haemostasis. Thrombin binds to GPIbα, triggering the signal transduction of platelet activation.[24] Hence, we assessed the effect of HY023016 on thrombin-mediated platelet activation. P-selectin is an adhesion molecule normally stored in a granule of platelets.[25] After platelet activity, P-selectin translocates with the granule to the surface and mediates platelet-leukocyte adhesion.[26,27] Similarly, GPIIb/IIIa mediates the common pathway of platelet activation-platelet aggregation. The antibody PAC-1 is an activation-specific monoclonal antibody (MoAb) directed against GPIIb/IIIa that does not bind to the unactivated GPIIb/IIIa.[28] Consequently, we investigated expression of CD62P and PAC-1, the indicated markers of platelet activation, via flow cytometry. Consistent with our hypothesis, HY023016 reduced platelet activation in a concentration-related manner, clarifying that HY023016 is one of potent inhibitors of thrombin-induced platelet activation.

Arteriovenous shunt thrombosis model is sensitive to both the modulation of the coagulation system and platelet function[29] and is widely used to evaluate antithrombotic drugs. Using intravital microscopy, we demonstrated that HY023016 dose-dependently inhibited thrombosis development in rat arteriovenous shunt thrombosis model. We also observed a decreased trend of embolus deposition

14401681, 2022, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/1440-1681.13634 by Harvard University, Wiley Online Library on [09/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

at silk thread at the early stages of thrombus formation. It may result from interference of HY023016 with GPIbα binding to vWF, which is a critical step in initial thrombosis. Additionally, HY023016 is expected to prolong both PT and aPTT, suggesting that HY023016 may inhibit the convergence point of the intrinsic and extrinsic coagulation pathways. HY023016 exhibited a slightly weaker or similar anti-thrombotic efficacy as compared to dabigatran etexilate and tetramethylpyrazine under the same conditions in the classical arteriovenous shunt thrombosis model. Moreover, bleeding is the major limitation of antithrombotics. We have confirmed that HY023016 did not significantly prolong tail bleeding time in mice.[10]

Together, these lines of evidences strongly indicate that HY023016 may represent a novel and potent antithrombotic agent by 'additive anticoagulant and antiplatelet effects'. Indeed, as a prodrug, pharmacokinetic studies of HY023016 deserve further investigation, and chemical structure requires closer optimization. However, it cannot be ignored that HY023016 might be a promising lead compound for development of novel antithrombotic drugs. Compared to traditional single-target antithrombotic drugs, HY023016 has similar or even better efficacy and a wider range of applications.

4 | MATERIALS AND METHODS

4.1 | Reagents and chemicals

HY023016, dabigatran etexilate (PubChem ID: 6445226) and tetramethylpyrazine (PubChem ID: 14296) were obtained from the Department of Medicinal Chemistry, China Pharmaceutical University (Purity > 98%). Human α-thrombin (Purity > 95%) was purchased from Haematologic Technologies. Sodium carboxymethyl cellulose (CMC-Na) and dimethyl-sulfoxide (DMSO) were purchased from Shanghai Reagent Station. S-2238 was purchased from Beijing Adhoc International Technologies. Amine coupling kit, CM5 sensor chip and HBS-EP buffer were purchased from Biacore AB. Fluorescein isothiocyanate (FITC) anti-human CD42b antibody was purchased from BioLegend. PE anti-human CD61 antibody, FITC anti-human CD62P antibody and anti-human PAC-1 antibody was purchased from eBioscience. PPACK was purchased from Cayman. Fluorescein isothiocyanate fibrinogen γ'-peptide was synthesized by Shanghai Top-peptide Bio. Calcein-AM was purchased from AAT Bioquest. Activated partial thromboplastin time assay kit and PT assay kit were purchased from Nanjing Jiangcheng Bioengineering Institute.

4.2 | Ethics statement

All procedures were conducted in accordance with the Declaration of Helsinki for human and the Guidelines on the Care and Use of Laboratory Animals for animals (Chinese Council on Animal Research and the Guidelines of Animal Care). Blood was donated by healthy volunteers for these experiments, who have signed an informed consent. The study was approved by the Ethical Committee of China Pharmaceutical University.

4.3 | Animals

Sprague-Dawley rats (180-220 g) were provided by the Laboratory Animal Center of China Pharmaceutical University. All animals were in well-ventilated animal house kept at 25 ± 5°C under 12 hour light and dark cycle. The animals were allowed to readjust to the new housing environment for 1 week before the experiment.

4.4 | Preparation of rat liver microsomes and treatments of drugs

After Sprague-Dawley rats were killed, the liver with cold HEPES buffer containing KCl (0.15 M) through the portal vein and rinse was perfused repeatedly until it became light yellow. Their livers were quickly removed. The livers were sliced and thoroughly homogenized with three-volumes of HEPES buffer using an FSH-2A homogenizer (DaSheng equipment), and centrifuged at 9000 *g* for 20 minutes at 4°C using 5810R centrifuge (Eppendorf) to produce a supernatant. The supernatant was transferred to ultracentrifuge tubes and then centrifuged at 100 000 *g* for 60 minutes at 4°C using an L-80XP ultracentrifuge (Beckman instrument). The microsomal pellets were resuspended in HEPES buffer with 30% glycerine. The amount of microsomal protein was determined by bicinchoninic acid (BCA) assay reagent.

HY023016, dabigatran etexilate and etramethylpyrazine was mixed with the rat liver microsome HEPES solution with a content of 1 mg/mL. After 4 hours of incubation of the mixture at 37°C, we added 1.725 mL of HEPES to it to obtain a drug-HEPES solution with a concentration of 10^{-3} M. We used 100 000 *g* ultra-high-speed centrifugation for 60 minutes to remove liver microsomes. Finally, the solution of drugs was diluted with HEPES buffer to series concentration.

4.5 | Preparation of human platelet-rich plasma and gel-filtered platelets

Human blood was collected from healthy volunteers (not taking drugs or smoking for at least 14 days), who have signed an informed consent. These experiments were approved by the Ethical Committee of China Pharmaceutical University. Venous blood containing 0.1 volume of ACD (71.38 mM citric acid, 85 mM sodium citrate and 27.78 mM glucose) was placed into siliconized vacutainer tubes. Platelet-rich plasma (PRP) was isolated from ACD blood at 230 g for 20 minutes. Gel-filtered platelets were isolated from PRP using a Sepharose 2B chromatography column with Tyrode buffer (138 mM NaCl, 2.7 mM KCl, 2 mM $MgCl_2$, 0.42 mM NaH_2PO_4, 5 mM

14401681, 2022, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/1440-1681.13634 by Harvard University, Wiley Online Library on [09/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

glucose, 10 mM HEPES, pH 7.4 and 0.2% bovine serum albumin). Platelet count was adjusted to 2×10^8 platelets $\times$ mL^{-1}.

4.6 | Surface plasmon resonance

Surface plasmon resonance measurements were carried out on a Biacore 3000 instrument. By the use of the amine coupling kit, the purified α-thrombin was covalently immobilized at pH 5.5 on a CM5 sensor chip, until ~1500 response units (RU) were reached. Aliquots of GPIbα N terminal 1-300 aa up to 16 μg $\times$ mL^{-1} were injected into the flow cell. The RU at steady state was fitted to the Langmuir equation to yield the K_d of the thrombin-GPIbα interaction and calculated other kinetic parameters. To measure the effect of test articles on this interaction, 50 μg $\times$ mL^{-1} GPIbα N terminal 1-300 aa was injected in the presence of 0-10,000 nM HY023016, dabigatran etexilate or tetramethylpyrazine. All measurements were carried out at 25°C in HBS-EP buffer at a flow rate of 30 μL/min^{-1}. IC50 values, used for characterizing the concentration of an inhibitor that was required for 50% binding inhibition of drugs on thrombin and GPIbα in vitro, were determined by plotting Req values and using non-linear regression to fit them.

4.7 | Flow cytometry

The interaction of HY023016 with different receptors expressed on the surface of platelets was tested as previously described.[30] The washed platelets were incubated with vehicle or different concentrations of HY023016 at 37°C for 30 minutes. Next, 5 μL of monoclonal antibody was added into above sample (100 μL). The expression of platelet membrane glycoproteins was investigated by using FITC-conjugated monoclonal antibodies anti-human GPIbα (CD42b), P-selectin (CD62P) and a PE conjugated monoclonal antibody anti-human GPIIIa (CD61). After 15 minutes of incubation at room temperature in the dark, the reaction was stopped by addition of 1% paraformaldehyde. Flow cytometry was performed with MACSQuant.

The effect of HY023016 on platelet activation was also assessed by expression of the indicated markers. Briefly, the washed platelets were incubated with different concentrations of HY023016, dabigatran etexilate or tetramethylpyrazine at 37°C for 30 minutes. 95 μL of the above-prepared sample was taken and 5 μL of α-thrombin was added to the sample to a final concentration of 10 nM. This mixture was then incubated for 10 minutes at room temperature, and subsequently was stained with 5 μL FITC-conjugated monoclonal antibodies anti-human PAC-1 or anti-human P-selectin, respectively. At last, the expressions of PAC-1 or P-selectin were analysed as described above.

We use Flowjo to analyse data from flow cytometry. In this software we can draw the positive fluorescence intensity range gate in histogram, which will give us a percentage that characterized as the number of cells in this gate. The gate of every group is identical.

4.8 | Fluorescent assay

To measure the effect of HY023016 on thrombin exosite II, we used PPACK to inhibit thrombin activity site by S-2238 hydrolysis. Previously we synthesized FITC-labelled fibrinogen γ'-peptide (^{408}VRPEHPAETE-Y(PO3)-DSL-Y(PO3)-PEDDL^{427}) by a solid-phase method.

Active-site-blocked thrombin and different concentrations of HY023016 (up to 4000 μM) were incubated for 15 minutes, and the final concentration of thrombin was 333 nM. The reaction mixture was added into a 2-mL cuvette containing 50 nM FITC-γ'-peptide. After 30 seconds of incubation, a fluorescence emission spectrum was obtained at 493 nm excitation wavelength with slits set at 3 nm for excitation and 5 nm for emission. The fluorescence intensity change was measured to calculate IC_{50} value by plotting inhibition vs HY023016 concentration.

$$\text{Inhibition rate} = \frac{\text{Administration group fluorescence value} - \text{Model fluorescence value}}{\text{Blank fluorescence value} - \text{Model fluorescence value}} \times 100\%$$

Moreover, to ensure the inhibition of HY023016 on thrombin binding to fibrinogen γ'-peptide, we also evaluated the Kd value in the presence or absence of HY023016. Briefly, PPACK-thrombin was added to the cuvette containing 45 nM fluorescein-γ'-peptide and 400 nM HY023016. Titration was continued up to 2.8 μM thrombin. As a control, the fluorescence intensity was measured without HY023016. In addition, we tested the spectrum of the Arg93 mutation thrombin variant, R93E, which can simulate the binding of γ' peptide to the inhibition of the exosite II site to some extent. Fluorescence parameters were set as described above. Dissociation constants were calculated by plotting F/F0 vs thrombin concentration curve and fitting the resulting curve to a one-site specific binding equation, where F0 was maximum fluorescence intensity of free peptide, and F was the fluorescence intensity of mixing fluorescence at 528 nm.

All experiments were performed in buffer containing 50 mM Tris-HCl, pH 7.4, 50 mM NaCl with 0.1% PEG8000 on HITACHI F-4600 fluorometer.

4.9 | Intravital microscopy of rat arteriovenous shunt thrombosis model

As previously described,[31,32] intravital microscopy was used to assess the thrombotic response to arteriovenous shunt thrombosis in rats with some modification. Blood was collected from carotid artery of male rats with chloral hydrate anaesthesia and was placed into siliconized tubes containing 0.1 volume of ACD. Next, PRP was isolated by centrifugation at 300 $\times$ *g* for 10 minutes. Gel-filtered platelets were obtained from PRP using a Sepharose 2B agarose gel column with PIPES buffer (5 mM PIPES, 137 mM NaCl, 4 mM KCl, 0.1% glucose, pH 7.0), and was adjusted to 2×10^7 platelets $\times$ mL^{-1}. Gel-filtered platelets (1 mL) was labelled with calcein acetoxymethyl ester (40 μg/mL) at 37°C for 15 minutes. HY023016 (2.5, 5

14401681, 2022, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/1440-1681.13634 by Harvard University, Wiley Online Library on [09/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

and 10 mg/kg), dabigatran etexilate (5 mg/kg), tetramethylpyrazine (5 mg/kg) and vehicle control (0.5% CMC-Na) were administered via oral gavage 60 minutes before anesthetizing the rats (six rats per group). Arteriovenous shunt thrombosis model was conducted as previously described.[33] A total of 300 μL of labelled platelets were injected into rats via the lateral tail vein. The middle Tygon tubing was chosen and visualized with a Zeiss LSM 710 fluorescent microscope (objective: ×200). We recorded for 15 minutes and quantified (a) the time required for the formation of the first 20 μm thrombus and (b) the number of thrombi of diameter >20 μm at the end of the observation. The shunt was then removed and a 5-cm length thrombus was withdrawn and immediately weighed. Subsequently, we collected rat blood and measured APTT and PT.

4.10 | Statistical analysis

All data are expressed as mean ± standard deviation (SD). Differences between the groups were analysed by one-way ANOVA followed by a Student Newman-Keuls test using GraphPad Prism version 6.0 unless otherwise stated. $P < 0.05$ was considered statistically significant.

ACKNOWLEDGEMENTS

This work was supported by grants from the National Natural science Foundation of China (grant numbers 81373415, 21272277).

CONFLICT OF INTEREST

The authors declare that there is no conflict of interest.

AUTHOR CONTRIBUTIONS

G.Q.G and Y.G.X. conceived and designed the research. G.Q.G., Y.G.X, and S.W. conducted the experiments. S.W., S.C., M.R.J., Q.W. and Y.Y.J. analysed the data. G.Q.G. and Y.G.X. wrote the manuscript.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author, GQ Gong, on reasonable request.

REFERENCES

1. Eisert WG, Hauel N, Stangier J, et al. Dabigatran: an oral novel potent reversible nonpeptide inhibitor of thrombin. *Arterioscler Thromb Vasc Biol*. 2010;30(10):1885-1889.
2. Xu XR, Carrim N, Neves MAD, et al. Platelets and platelet adhesion molecules: novel mechanisms of thrombosis and anti-thrombotic therapies. *Thromb J*. 2016;14(1):37-46.
3. Li M, Handa S, Ikeda Y, et al. Specific inhibiting characteristics of tetramethylpyrazine, one of the active ingredients of the Chinese herbal medicine 'Chuanxiong', on platelet thrombus formation under high shear rates. *Thromb Res*. 2001;104(1):15-28.
4. Yang WH, Gong GQ, Zhou Y, et al. Effect and mechanism of tetramethylpyrazine on antithrombotic. *Chin J Clin Pharmacol Therapeut*. 2012;17(3):241-245.
5. Wu GX, Wu JC, Ma HT, et al. Inhibitory effects of tetramethylpyrazine on platelets during cardiopulmonary bypass and arterial thrombus formation in dogs. *Zhongguo Yao Li Xue Bao = Acta Pharmacol Sin*. 1992;13(4):330.
6. Ezekowitz MD, Reilly PA, Nehmiz G, et al. Dabigatran with or without concomitant aspirin compared with warfarin alone in patients with nonvalvular atrial fibrillation (PETRO Study). *Am J Cardiol*. 2007;100(9):1419-1426.
7. Michelson AD. Advances in antiplatelet therapy. *Hematology Am Soc Hematol Educ Program*. 2011;2011(1):62-69.
8. Leiba M, Malkiel S, Budnik I, et al. Thrombin generation as a predictor of thromboembolic events in multiple myeloma patients. *Blood Cells Mol Dis*. 2017;65:1-7. 10.1016/j.bcmd.2017.03.010. PubMed PMID: 28365523; eng.
9. Sidhu PS, Aziz MHA, Sarkar A, et al. Designing allosteric regulators of thrombin. exosite 2 features multiple sub-sites that can be targeted by sulfated small molecules for inducing inhibition. *J Med Chem*. 2013;56(12):5059-5070.
10. Li YZ, Gong GQ, Yang WH, et al. Antithrombotic activity of HY023016, a novel dabigatran prodrug evaluated in animal thrombosis models. *Thromb Res*. 2013;131(5):425-435.
11. Da Q, Teruya M, Guchhait P, et al. Free hemoglobin increases von Willebrand factor-mediated platelet adhesion in vitro: implications for circulatory devices. *Blood*. 2015;126(20):2338-2341.
12. Huizinga EG, Tsuji S, Romijn RA, et al. Structures of Glycoprotein Ibα and Its Complex with von Willebrand Factor A1 Domain. *Science*. 2002;297(5584):1176–1179. http://doi.org/10.1126/science.107355
13. Celikel R, Mcclintock RA, Roberts JR, et al. Modulation of α-thrombin function by distinct interactions with platelet glycoprotein Ibα. *Science*. 2003;301(5630):218.
14. Candia ED. Mechanisms of platelet activation by thrombin: a short history ☆. *Thromb Res*. 2012;129(3):250-256.
15. Fujimoto M, Salamon N, Mayor F, et al. Characterization of arterial thrombus composition by magnetic resonance imaging in a swine stroke model. *Stroke*. 2013;44(5):1463-1465.
16. Goldhaber SZ. Venous thromboembolism: epidemiology and magnitude of the problem. *Best Pract Res Clin Haematol*. 2012;25(3):235-242.
17. Dong JF, Li CQ, López JA. Tyrosine sulfation of the glycoprotein Ib-IX complex: identification of sulfated residues and effect on ligand binding. *Biochemistry*. 1994;33(46):13946-13953.
18. Marchese P, Murata M, Mazzucato M, et al. Identification of three tyrosine residues of glycoprotein Ib alpha with distinct roles in von Willebrand factor and alpha-thrombin binding. *J Biol Chem*. 1995;270(16):9571-9578.
19. Bock PE, Panizzi P, Verhamme IM. Exosites in the substrate specificity of blood coagulation reactions. *J Thromb Haemost*. 2007;5(Supplement s1):81-94.
20. Krishnaswamy S. The transition of prothrombin to thrombin. *J Thromb Haemost*. 2013;11(s1):265-276.
21. Andrews RK, Gardiner EE, Shen Y, et al. Glycoprotein Ib-IX-V. *Int J Biochem Cell Biol*. 2003;35(8):1170.
22. Treuheit NA, Beach MA, Komives EA. Thermodynamic compensation upon binding to exosite 1 and the active site of thrombin. *Biochemistry*. 2011;50(21):4590.
23. Malovichko MV, Sabo TM, Maurer MC. Ligand binding to anion-binding exosites regulates conformational properties of thrombin. *J Biol Chem*. 2013;288(12):8667-8678.
24. Soslau G, Class R, Morgan DA, et al. Unique pathway of thrombin-induced platelet aggregation mediated by glycoprotein Ib *. *J Biol Chem*. 2001;276(24):21173-21183.
25. Mcever RP. Adhesive interactions of leukocytes, platelets, and the vessel wall during hemostasis and inflammation. *Thromb Haemost*. 2001;86(3):746.
26. Frenette PS, Moyna C, Hartwell DW, et al. Platelet-endothelial interactions in inflamed mesenteric venules. *Blood*. 1998;91(4):1318-1324.

14401681, 2022, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/1440-1681.13634 by Harvard University, Wiley Online Library on [09/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

27. Larsen E, Celi A, Gilbert GE, et al. PADGEM protein: a receptor that mediates the interaction of activated platelets with neutrophils and monocytes. *Cell*. 1989;59(2):305-312.
28. Peter K, Schwarz M, Ylanne J, et al. Induction of fibrinogen binding and platelet aggregation as a potential intrinsic property of various glycoprotein IIb/IIIa (alphaIIbbeta3) inhibitors. *Blood*. 1998;92(9):3240-3249. PubMed PMID: 9787160; eng.
29. Letienne R, Leparq PAA, Calmettes Y, et al. PAR1 antagonist mediated antithrombotic activity in extracorporeal arterio-venous shunt in the rat. *Thromb Res*. 2010;125(3):257.
30. Ferreira MAD, Nascimento NRFD, Sousa CMD, et al. Oncocalyxone A inhibits human platelet aggregation by increasing cGMP and by binding to GP Ibα glycoprotein. *Br J Pharmacol*. 2008;154(6):1216-1224.
31. Banno F, Chauhan AK, Kokame K, et al. The distal carboxyl-terminal domains of ADAMTS13 are required for regulation of in vivo thrombus formation. *Blood*. 2009;113(21):5323-5329.
32. Yang Y, Shi Z, Reheman A, et al. Plant food delphinidin-3-glucoside significantly inhibits platelet activation and thrombosis: novel protective roles against cardiovascular diseases. *PLoS One*. 2012;7(5):e37323.
33. Umar A, Yimin W, Tohti I, et al. Effect of traditional Uyghur medicine abnormal Savda Munziq extract on rabbit platelet aggregation in vitro and rat arteriovenous shunt thrombosis in vivo. *J Ethnopharmacol*. 2015;159:184-188.

SUPPORTING INFORMATION

Additional supporting information may be found in the online version of the article at the publisher's website.

How to cite this article: Wan S, Cui S, Jiang M, et al. Dual-target synergistic antithrombotic mechanism of a dabigatran etexilate analogue (HY023016). *Clin Exp Pharmacol Physiol*. 2022;49:567–576. doi:10.1111/1440-1681.13634

14401681, 2022, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/1440-1681.13634 by Harvard University, Wiley Online Library on [09/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License